Queen Mary
University of London
Science and Engineering

EBU4202: Digital Circuit Design
Tutorials (Block 4)

Dr. Md Hasanuzzaman Sagor (Hasan)
Dr. Chao Shu (Chao)
Dr. Farha Lakhani (Farha)
Dr. Fatma Benkhelifa (Fatma)

School of Electronic Engineering and Computer Science,
Queen Mary University of London,
London, United Kingdom.

Question 1 & 2

- Explain what is meant by each of the following terms:

  i) Non-volatile

  ii) Random Access

  iii) DRAM

  iv) EPROM

Q2. Using a suitable diagram, briefly explain the function of a 2-input selector/MUX.

Question 3

- Suppose we have a computer with 256 MBytes of memory, with each memory chip having a capacity of 4 MBytes.

  i) How many memory chips are required?

  ii) How many address bits are required?

  iii) How many bits are required to select the memory chips?

  iv) Draw a block diagram of a circuit using a decoder so that all 256 Mbytes of memory can be addressed.

Question 4

Design a 16 x 1-bit read-only memory (ROM) circuit pre-programmed with the data shown in the following table.

Address	Data
0	0
1	1
2	0
3	0
4	1
5	1
6	1
7	1
8	1
9	0
10	0
11	1
12	0
13	0
14	1
15	1

Question 5

Design an 8 x 4-bit read-only memory (ROM) circuit pre-programmed with the data shown in the Table below. (Hint: diodes, decoder, multiplexer etc. are required in this ROM design).

Address	Data (4 bit)
0	4
1	9
2	10
3	12
4	15
5	1
6	3
7	0

Question 6

Draw the Block Diagram of a Microcomputer showing CPU, memory and the three busses.